Savitribai Phule Pune University

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Department Of Computer Engineering

A.Y. [2024-25]

Library Management System

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CERTIFICATE

This is to Certify that Group No Division A Branch Computer Engineering has successfully completed the work associated with **Database Management System(310241)** titled as Library Management System as prescribed in the SPPU curriculum.

Place: Nashik
Date :

Ms.A.Y.Panpatil	Dr.D.V.Patil	Dr. Sane
Project Guide	**HOD**	**Principal**

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ABSTRACT

The Library Management System (LMS) is an application designed to streamline the processes of managing library resources, including books, periodicals, and digital media. Utilizing a Database Management System (DBMS), the LMS facilitates efficient data storage, retrieval, and manipulation. This system supports functionalities such as cataloging, borrowing, returning, and searching for library items, ensuring optimal resource utilization.

The architecture consists of a user-friendly interface for both librarians and patrons, enhancing user experience while providing robust security features to protect sensitive data. The backend employs relational database models to maintain integrity and support complex queries, allowing for detailed reporting and analytics. By automating routine tasks, the LMS improves operational efficiency, reduces errors, and fosters a more accessible and organized library environment. This abstract outlines the necessity for a comprehensive LMS, its core features, and the advantages of implementing a DBMS-driven solution in modern libraries.

Chapter 1: Introduction

The system features two distinct portals: the Admin Portal and the User Portal. The Admin Portal is designed for librarians and staff, allowing them to manage library inventory, user registrations, and transactions efficiently. Administrators can catalog new items, track current holdings, generate reports, and oversee all library operations, ensuring smooth functioning and optimal resource allocation.

Conversely, the User Portal provides library patrons with an intuitive interface to explore and access library resources. Users can search for books, check availability, borrow items, and manage their accounts, including viewing borrowing history and due dates. The User Portal empowers patrons by making library services more accessible and user-friendly.

By utilizing a DBMS, the LMS ensures data integrity, security, and efficient retrieval of information. This dual-portal system not only enhances operational efficiency but also fosters a more engaging and accessible library environment for all users. The LMS represents a crucial advancement in modern library management, catering to the diverse needs of both administrators and patrons.

1.1 Problem Statement

To design and implement a comprehensive Library Management System (LMS) that efficiently manages the operations of a library, including the tracking of books, members, transactions, and other related activities, thereby enhancing user experience and streamlining library processes. Admin can Manage book inventory, member registrations, and oversee transactions.
Student or memeber can search for books, borrow/return items, and view account status.

1.2 Objectives

1. User Management: Allow administrators to add, edit, and remove users (students, faculty, etc.) and manage their roles and permissions.

2. Inventory Management: Enable the addition, deletion, and modification of library resources (books, journals, etc.) with detailed metadata.

3. Account Management: Enable users to create and manage their profiles, including personal information and borrowing history.

4. Search Functionality: Provide an intuitive search feature for users to find library resources by title, author, or category.

5. Borrowing and Returning: Allow users to borrow and return items online, view due dates, and renew loans if necessary.

1.3 Scope of Project

1.Resource Management:

Cataloging and managing various types of library resources, including books, journals, and digital media. Tracking item availability and status (e.g., checked out, reserved, lost).

2. User Management:

Enabling user registration and profile management for library patrons.
Providing features for tracking user borrowing history and managing fines.

3. Admin Portal Features:

Allowing librarians to add, update, or delete library resources.
Generating reports on inventory status, user activities, and circulation statistics.
Managing user accounts and permissions, ensuring secure access to the system.

4. User Portal features:

Allowing users to view their account details, including current loans, due dates, and overdue items.Enabling users to receive notifications about new arrivals and reminders for due dates.

5. Transaction Management:

Automating borrowing and returning processes to minimize manual intervention.
Maintaining accurate transaction records for auditing and reporting purposes.

6. Search and Retrieval:

Implementing advanced search functionality for users to find resources by title, author, subject, or ISBN.Providing filters and sorting options to enhance the user experience.

Chapter 2: Literature Survey

The integration of a DBMS is crucial for ensuring data integrity, security, and efficient data handling. Research by Mishra et al. (2019) emphasizes the importance of using relational databases to manage complex queries and maintain data relationships effectively. The use of SQL databases allows for efficient searching, retrieval, and updating of records, enhancing the overall performance of the LMS.

User experience plays a significant role in the success of an LMS. Studies by Hernandez et al. (2020) demonstrate that intuitive interfaces in user portals lead to increased user satisfaction and engagement. Features such as advanced search options, personalized dashboards, and mobile access are increasingly prioritized to meet user expectations.

The admin portal is essential for managing the operational aspects of the library. Research by Singh and Kumar (2021) highlights key functionalities such as inventory management, user account administration, and reporting tools. These features empower librarians to efficiently handle daily operations and make data-driven decisions.

Chapter 3: System Requirement Specification

3.1 Hardware Requirement:

4 GB RAM
64 GB HDD

3.2 Software Requirement:

OS: Windows / Ubuntu

Basic Internet Connection

- Visual Studio Code
- XAMPP
- MYSQL

3.3 Language used :

1. Frontend : Html, Css and Javascript.

2. Backend : mysql and php.

Chapter 4 : System Design

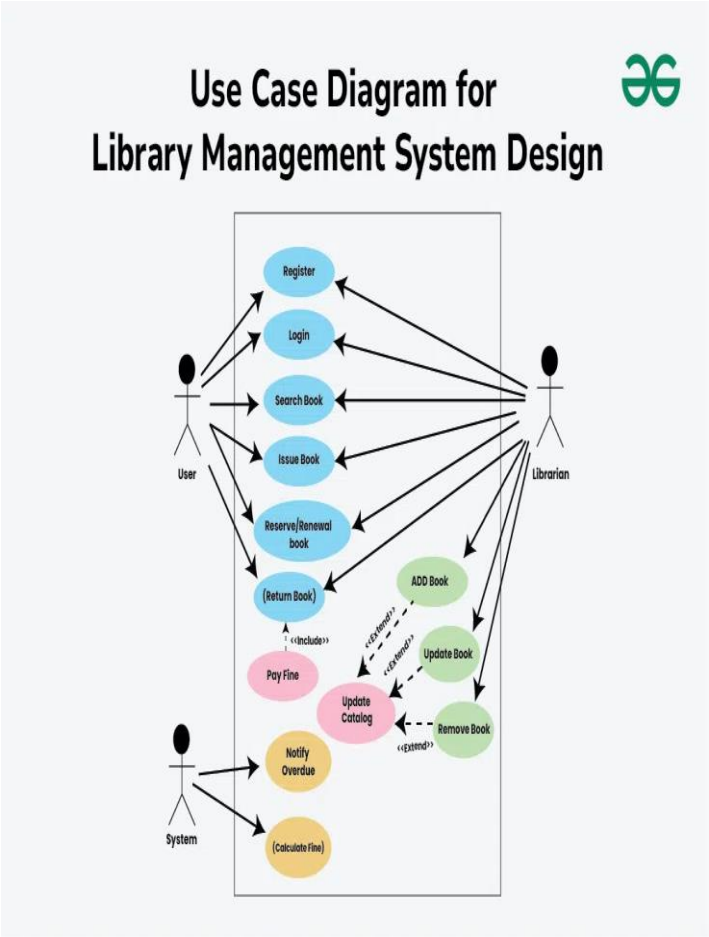
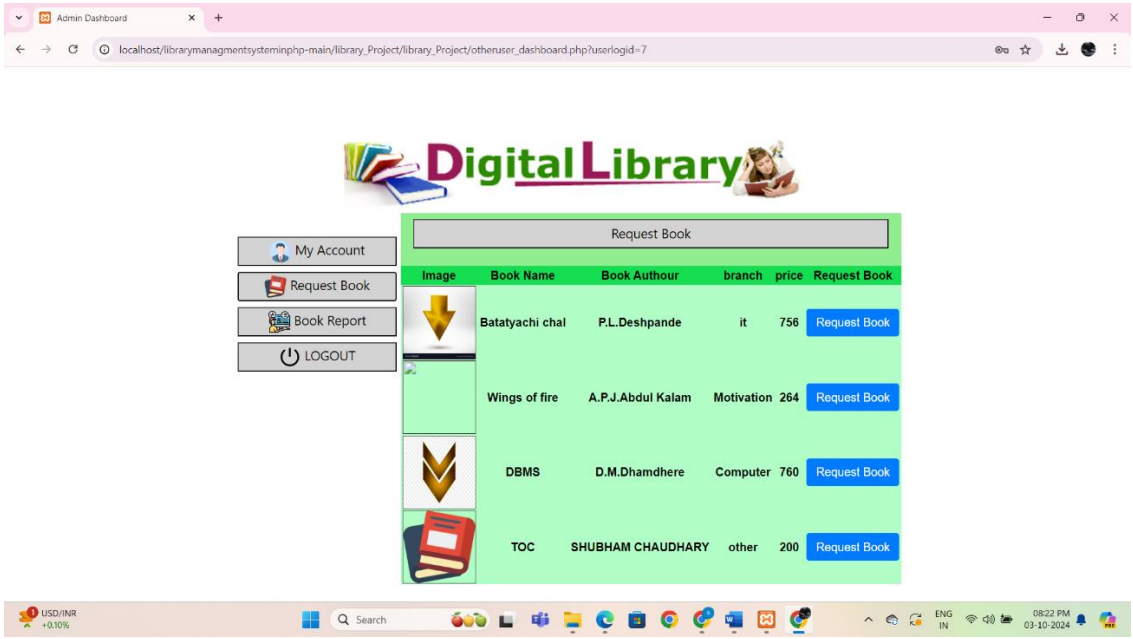
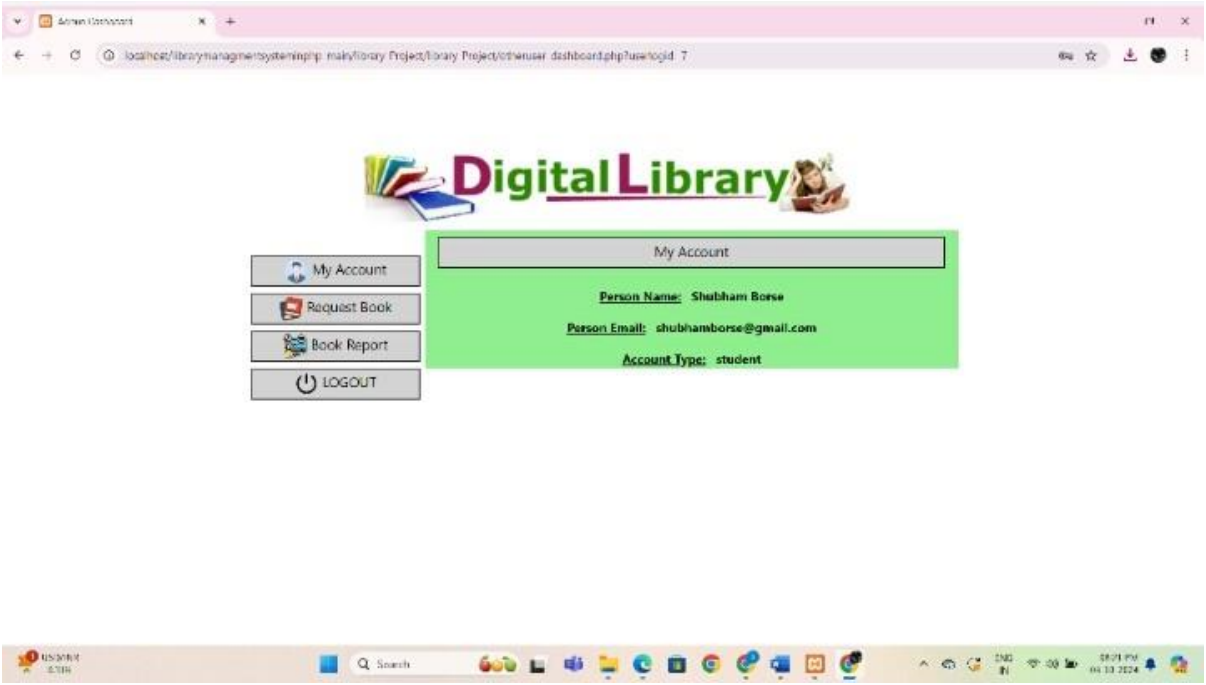


fig. Library management system

Chapter 5 : Result and Disccusion

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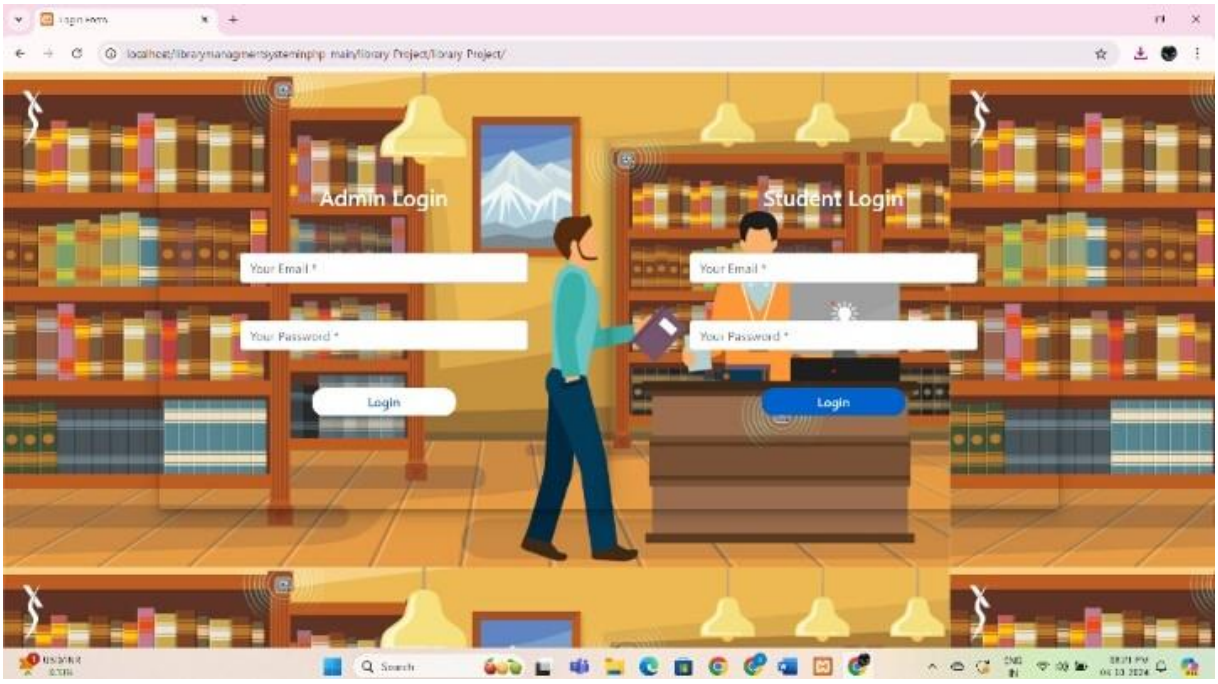
Book Report
LOGOUT
My Account
Person Name: Shubham Borse
Person Email: shubhamborse@gmail.com
Account Type: student


Admin Login
Your Email *
Your Password *
Login
Student Login
Your Email *
Your Password *
Login

Chapter 6 : Conclusion

The Library Management System (LMS) designed with distinct admin and user portals effectively streamlines library operations and enhances user experience. The admin portal facilitates efficient management of library resources, enabling administrators to oversee book inventory, manage user accounts, and generate reports with ease. Meanwhile, the user portal empowers patrons to search for books, manage their accounts, and access resources seamlessly. This dual-portal approach not only improves operational efficiency but also fosters a user-friendly environment, ultimately promoting better engagement with library services. By implementing a robust database management system, the LMS ensures data integrity, security, and scalability, positioning the library to adapt to future needs and advancements in technology.

Future Scope :

1. Integration with Digital Libraries: Expanding access to e-books and online resources, allowing users to borrow digital content seamlessly.

2. Mobile Applications: Developing mobile apps for both users and administrators to enhance accessibility and convenience in managing library resources on the go.

3. Artificial Intelligence: Implementing AI-driven recommendations and chatbots to assist users in finding books and answering queries efficiently.

4. Data Analytics: Utilizing data analytics to track user behavior, optimize resource allocation, and improve services based on user preferences and trends.

5. Improved User Engagement: Introducing features like personalized reading lists, notifications for new arrivals, and community features for user interactions and reviews.